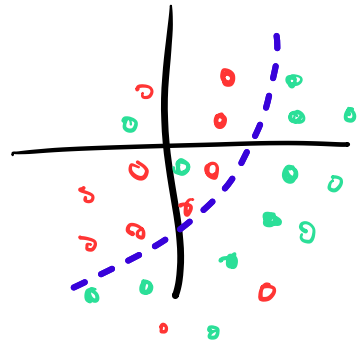


Week-11 Part A: Soft-Margin SVM

$$\min_{w \in \mathbb{R}^d} \quad \frac{1}{2} \|w\|^2 + C \cdot \sum_{i=1}^n \varepsilon_i$$

$$\text{such that: } (w^T x_i) y_i + \varepsilon_i \geq 1 \quad \forall i$$
$$\varepsilon_i \geq 0 \quad \forall i$$

Can this be kernelized?



$$L(w, \varepsilon, \alpha, \beta) = \frac{1}{2} \|w\|^2 + C \cdot \sum_{i=1}^n \varepsilon_i + \sum_{i=1}^n \alpha_i (1 - w^T x_i y_i - \varepsilon_i) + \sum_{i=1}^n \beta_i (-\varepsilon_i)$$

Dual Problem

Primal

$$\min_{w, \varepsilon} \left[\max_{\substack{\alpha \geq 0 \\ \beta \geq 0}} \frac{1}{2} \|w\|^2 + C \cdot \sum_{i=1}^n \varepsilon_i + \sum_{i=1}^n \alpha_i (1 - w^T x_i y_i - \varepsilon_i) + \sum_{i=1}^n \beta_i (-\varepsilon_i) \right]$$

Dual

$$\max_{\substack{\alpha \geq 0 \\ \beta \geq 0}} \left[\min_{w, \varepsilon} \frac{1}{2} \|w\|^2 + C \cdot \sum_{i=1}^n \varepsilon_i + \sum_{i=1}^n \alpha_i (1 - w^T x_i y_i - \varepsilon_i) + \sum_{i=1}^n \beta_i (-\varepsilon_i) \right]$$

Fix some α & β

$$\frac{\partial L}{\partial w} = 0 \Rightarrow \boxed{w_{\alpha, \beta}^* = \sum_{i=1}^n \alpha_i x_i y_i} \quad (\text{ie}) \quad w_{\alpha, \beta}^* = X^T \alpha$$

$$\frac{\partial L}{\partial \varepsilon_i} = 0 \Rightarrow C - \alpha_i - \beta_i = 0 \Rightarrow \boxed{\alpha_i + \beta_i = C}$$

$$\max_{\substack{\alpha \geq 0 \\ \beta \geq 0 \\ \alpha + \beta = C \cdot 1}} \left[\alpha^T 1 - \alpha^T Y^T X^T X Y \alpha \right]$$

Exactly the same objective function

New constraints : $\beta_i \geq 0$ & $\alpha_i + \beta_i = C \quad \forall i$

ie $\alpha_i \leq C \quad \forall i$

$$\max_{\underline{0 \leq \alpha \leq C}} \left[\alpha^T 1 - \alpha^T Y^T \underline{X^T X} Y \alpha \right]$$

Only difference : α now has an upper bound

Box constraints

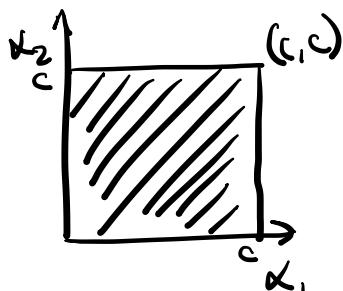
if $C = 0 \Rightarrow \alpha^* = 0 \in \mathbb{R}^n$

$$\therefore w^* = \sum_{i=1}^n \alpha_i^* x_i y_i \Rightarrow \boxed{w^* = 0}$$

ie. if $C = 0$, bribes don't cost anything (from primal)
linear combination (dual)

$C = \infty \Rightarrow$ No upper bound for α
(ie) same as hard margin

if penalty for bribe = ∞ , then only linearly separable datasets will work (from primal)



Kernelizable!

Is w^* still a sparse linear combination of datapoints?

Complementary Slackness conditions for Soft Margin SVM

(ie) link between primal optimum & dual optimum

$w^* \quad \varepsilon^*$ = primal optimum

$\alpha^* \quad \beta^*$ are dual optimum

$$1 : \forall i : \alpha_i^* (1 - w^{*T} x_i y_i - \varepsilon_i^*) = 0$$

$$2 : \forall i : \beta_i^* (\varepsilon_i^*) = 0$$

Case 1: $\alpha_i^* = 0$ Case 2: $\alpha_i^* \in (0, c)$ Case 3: $\alpha_i^* = c$

i) $\alpha_i^* = 0 \Rightarrow \beta_i^* = c$

$\therefore \varepsilon_i^* = 0$
 $\Rightarrow w^{*T} x_i y_i + \varepsilon_i \geq 1$

$\therefore w^{*T} x_i y_i \geq 1$

(ie that point doesn't need to pay a bribe - already correctly classified)
 it is either on the supporting hyperplanes or away from them (not within)

ii) $\alpha_i^* \in (0, c)$ ie $0 < \alpha_i^* < c$

$\therefore \beta_i^* \in (0, c)$ ie $0 < \beta_i^* < c$

$\therefore \varepsilon_i^* = 0$

$\therefore 1 - w^{*T} x_i y_i - \varepsilon_i^* = 0 \Rightarrow \underline{w^{*T} x_i y_i = 1}$

This point lies ON the supporting hyperplane
 $\Rightarrow$ These points contribute to w^* but with a value $< c$
 These are the support vectors

iii) $\alpha_i^* = c$ $\Rightarrow \beta_i^* = 0$

$1 - w^{*T} x_i y_i - \varepsilon_i^* = 0 \quad \therefore \varepsilon_i^* \geq 0$

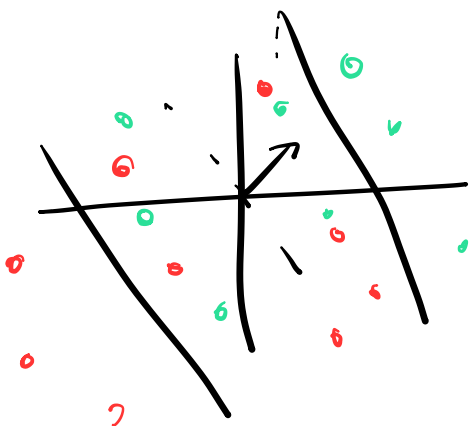
$\therefore 1 - w^{*T} x_i y_i \geq 0$

(ie) $\boxed{w^{*T} x_i y_i \leq 1}$

These points either lie on the supporting hyperplanes or on the wrong side of their respective hyperplane.

(could still be correctly classified but on wrong side of margin)

Most important to w^* , also support vectors



Anything for which $\alpha_i^* > 0$ is a support vector

Looking at this from primal's perspective

Case 1: $w^*{}^T x_i y_i < 1$ (either misclassified or classified correctly but not with sufficient margin)

$$w^*{}^T x_i y_i + \xi_i^* \geq 1 \Rightarrow \xi_i^* \geq 1 - w^*{}^T x_i y_i$$

$\therefore \xi_i^* > 0$ (need to pay a "bribe" to go to the margin)

$$\therefore \beta_i^* = 0 \Rightarrow \boxed{\alpha_i^* = C}$$

$\therefore$ if a point lies on the wrong side of the hyperplane, it is one of the most important points

Case 2: $w^*{}^T x_i y_i = 1$

$$\xi_i^* \geq 1 - w^*{}^T x_i y_i$$

$$\therefore \xi_i^* \geq 0 \Rightarrow \alpha_i^* \in [0, C]$$

Mathematically we can only say that $\xi_i^* \geq 0$ when we express this as a feasibility constraint but due to the minimization objective $\xi_i^* = 0$ always if x_i lies on the margin.

Case 3: $w^*{}^T x_i y_i > 1$

$$\Rightarrow 1 - w^*{}^T x_i y_i < 0$$

$$\therefore 1 - w^*{}^T x_i y_i - \xi_i^* < 0$$

$$\therefore \alpha_i^* = 0$$

points which are correctly classified with more than sufficient margin, then it does not matter for our algorithm

Summary

$$\alpha_i^* = 0 \Rightarrow w^*{}^T x_i y_i \geq 1$$

$$0 < \alpha_i^* < C \Rightarrow w^*{}^T x_i y_i = 1$$

$$\alpha_i^* = C \Rightarrow w^*{}^T x_i y_i \leq 1$$

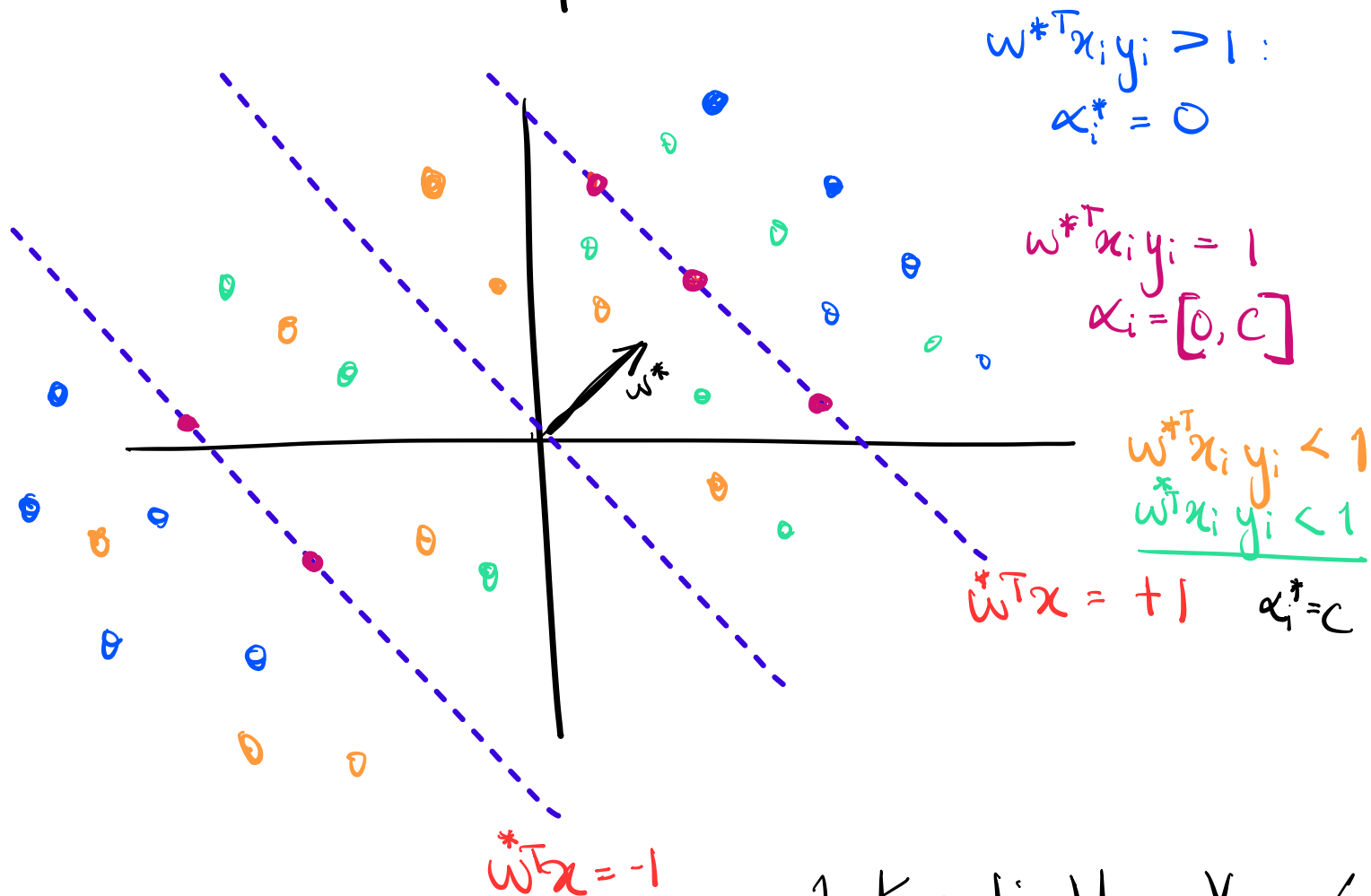
$$w^*{}^T x_i y_i < 1 \Rightarrow \alpha_i^* = C$$

$$w^*{}^T x_i y_i = 1 \Rightarrow 0 \leq \alpha_i^* \leq C$$

$$w^*{}^T x_i y_i > 1 \Rightarrow \alpha_i^* = 0$$

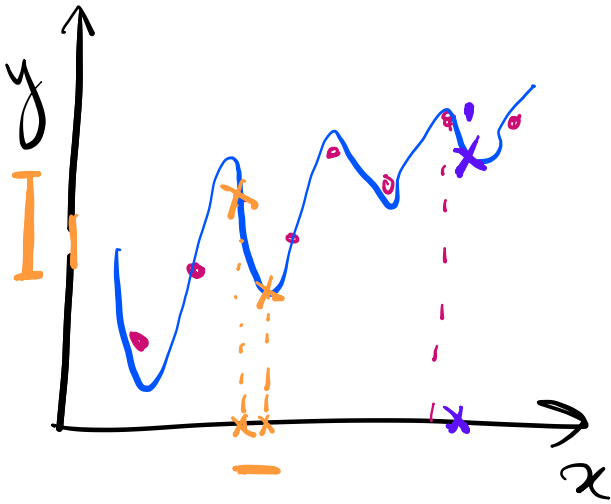
Dual: given importance (α 's) of datapoints $\Rightarrow$ what can I infer about location of datapoints

Primal: given location of datapoints $\Rightarrow$ what can I infer about their importance (to w^*)



1. Kernelizable - Yes ✓
2. Sparse solution - Yes ✓

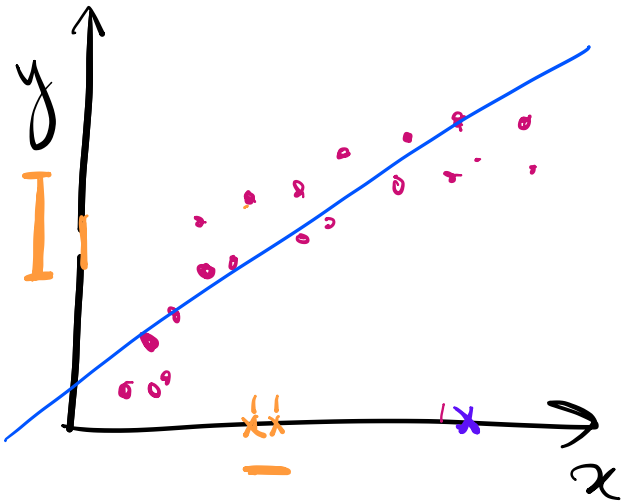
Underfitting & Overfitting



Training error = Low

Test error = high

Variance is too high!
Test error = high
 $\Rightarrow$ Overfitting



Train error = high

Test error = high

Underfitting

$E[\text{actual labels} - \text{predicted labels}]$ ← Bias = high
 These 2 "biases" are not the same

$w^T x + b$
 $\downarrow$ weight $\rightarrow$ bias

Ridge: $w \sim N(0, \sigma^2) \rightarrow$ Biasing your optimal w to be closer to zero
 $w \sim N(\mu, \sigma^2) \rightarrow$ Biasing w to be close to μ

$$\lambda \|w\|^2 \quad \times \quad \lambda \|\mu - w\|^2$$

Bias: On avg: how wrong is your prediction

Train error = low } Correctly captured
Test error = low } the relationship

Bias = low & variance = low Perfect Model X

Train error = high } fiction!
Test error = low } never happens
in reality

Analogy for Bias & Variance (with a Bulls-Eye)

